Name: _____ Mark: /26

Practical Assessment: Indicators from coloured plant material

This task is designed to test your practical skills as well as your knowledge of acids, bases and indicators.

Your Task

Your task is to create **two** simple acid-base indicators from red cabbage and other coloured plant material. You will then test your indicators with hydrochloric acid and sodium hydroxide solutions to determine their colour in acids and bases.

Equipment

- 100mL beaker x 2
- Distilled water
- Red cabbage
- Coloured plant material
- Bunsen burner
- Tripod
- Gauze mat
- Heatproof mat
- Wooden tongs
- Dimple tray
- Hydrochloric acid
- Sodium hydroxide
- Pipette x 2
- Safety glasses

Method

Part A: Making the indicators

1. Set up the tripod, Bunsen burner and gauze mat on the heatproof mat at your station.
2. Collect half a beaker of red cabbage.
3. Add just enough distilled water to cover the cabbage.
4. Heat the beaker until the water is boiling
5. Boil the water for 3-5 minutes or until the water is strongly coloured.
6. Remove from the heat using the tongs and allow it to cool.
7. Repeat the process for material from one other plant. Tear the material into small pieces before boiling.

Part B: Testing the indicators

1. Put two drops of hydrochloric acid into the dimple tray.
2. Using a pipette, add one or two drops of your indicator to the hydrochloric acid.
3. Record any colour change in the table over the page.
4. Put two drops of sodium hydroxide into the dimple tray.
5. Using a pipette, add one or two drops of your indicator to the sodium hydroxide.
6. Record any colour change in the table over the page.
7. Repeat for your other indicator using a clean pipette.

Part C: Testing Unknowns

1. Choose one of your indicators to use.
2. Put two drops of unknown A into a dimple on the dimple tray.

3. Test the unknown with your indicator and decide whether it is an acid, a base or neutral from the colour change and record your results over the page.
4. Repeat for the other three unknowns.

Results

/6

Plant	Colour in Neutral solution	Colour in Acidic Solution	Colour in Basic Solution
Red Cabbage			

Using your Results

Use one of your indicators to classify the unknown solutions as **acids**, **bases** or **neutral**. /4

Unknown A: _____ Unknown B: _____

Unknown C: _____ Unknown D: _____

Questions

1. What is an acid? /1

2. What is a base? /1

3. Give two properties of acids. /2

4. Give two properties of bases. /2

5. Hydrochloric acid is classed as a strong acid and acetic acid (vinegar) is a weak acid. Describe the difference between a strong acid and a weak acid. /2

6. A bull-ant bite hurts because the ant injects formic acid when they bite. Name a /2 household product which would stop the pain and give a reason why it would work.

Working in the Laboratory

Appropriate use of safety equipment
/2

Appropriate behaviour
/2

Workstation cleaned properly
/2